BloodLink

رابط الحياة

Project Documentation

1. Project Planning & Management
2. Literature Review
3. Requirements Gathering

Program	DEPI (Digital Egypt Pioneers Initiative), 2026
Track	Software Development - React Frontend Web Developer
Team	Ibrahim Hassan Anwar (Team Leader) Khaled Atef Salah Mohammed Eslam Mohammed Hager Waleed Shaker Zeyad Mohammed Mohammed Marawan Hesham Tolba
Repository	github.com/Marnie0/bloodlink
Date	16 October 2026

Phase 1: Project Planning & Management

Track: Software Development - React Frontend Web Developer\

1. Project Proposal

Overview

BloodLink is a community website that connects people in Egypt who need blood with suitable donors nearby. Every request is backed by an official hospital paper. Donor and patient always meet at the hospital, and no money is involved.

Problem

- Blood requests are spread through Facebook and WhatsApp posts that reach people by chance.
- Nobody can tell if a post is real or still needed.
- Most people who see a post have the wrong blood type or live too far away.
- Contact details are posted publicly, which invites scams and requests for money.

Solution

Step	What happens
Donor joins	Uploads an official hospital certificate showing their blood type
Patient asks	Uploads an official hospital paper proving the patient needs blood
Paper check	Partner hospital paper: confirmed by the hospital, badge **Verified**. Other official hospital: checked by AI, badge **Paper looks valid**.
Matching	Compatible, available donors closest to the hospital are notified first, and the area grows in waves
Accept	The donor accepts. Phone numbers are shared only then.
Meet	At the hospital. The hospital decides medically. Partner hospitals scan the donor's QR code to record the donation.

Objectives

1. Send each request only to donors who can actually help: compatible blood type, available, and nearby.
2. Show how much each request and donor can be trusted, using hospital papers and clear badges.
3. Protect users: phone numbers stay hidden until a donor accepts, meetings happen only at hospitals, and money requests are banned.
4. Deliver a working, deployed Arabic website that works well on phones and computers, by 30 Nov 2026.

Scope

In scope (core)	Out of scope
Accounts with 4 roles: donor, requester, partner hospital, admin	Examining donors or making any medical decision

In scope (core)	Out of scope
Certificate and hospital paper upload	Guaranteeing a paper is genuine
AI paper check and partner-hospital confirmation	Payments of any kind
Matching by blood type, distance and trust level	SMS or WhatsApp Business API
Web push notifications in waves	Mobile apps (a possible future improvement; this project is a website only)
Accept or decline, with phone numbers hidden until acceptance	
QR check-in at partner hospitals; self-confirmation elsewhere	
Live request progress	
Admin: paper review, reports, bans, key numbers	
Arabic RTL interface that works on phones	

Planned extras (built only after checkpoint 2): an English version, a map view, a WhatsApp share card, donation history and charts.

Main deliverables

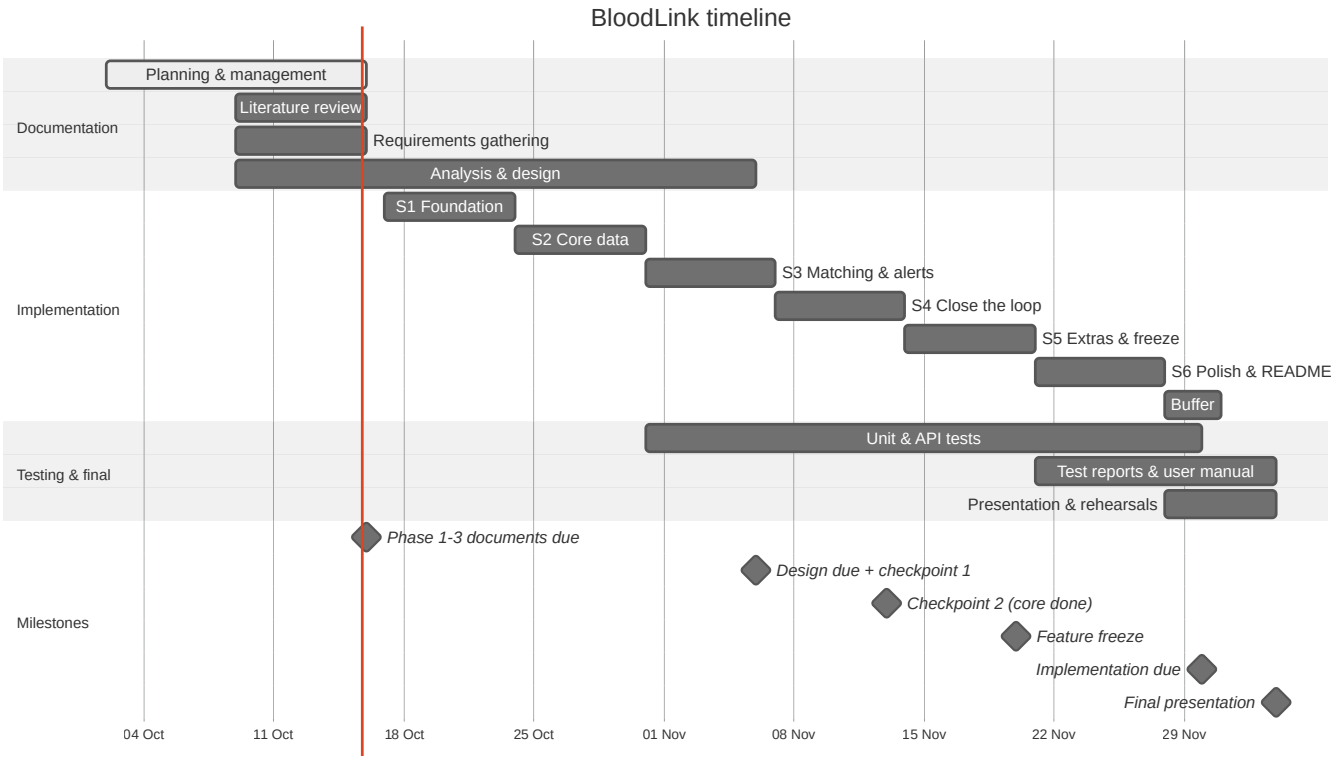
- Deployed website with demo data
- Source code on GitHub, with a README and API documentation
- Phase documents (planning, requirements, design, testing)
- Presentation, demo video and project infographic

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2. Project Plan

Timeline (Gantt chart)

The plan follows the DEPI deadlines. Implementation runs in six one-week sprints, Saturday to Friday, with a three-day buffer before the 30 Nov deadline.



Sprints

Sprint	Dates	Goal	Done when
S1 Foundation	17 to 23 Oct	Repo, TypeScript, Docker, deploy, data models, login with roles, Arabic UI kit	The website is deployed and a user can log in with a role
S2 Core data	24 to 30 Oct	Donor signup and certificate upload, request form and paper upload, partner confirmation	A request becomes active after the hospital confirms it
S3 Matching & alerts	31 Oct to 6 Nov	Matching with tests, AI paper check, push notifications, accept or decline, live progress	**Checkpoint 1:** the whole flow works roughly
S4 Close the loop	7 to 13 Nov	QR check-in at partner hospitals, self-confirmation elsewhere, waves, admin tools	**Checkpoint 2:** all core features are done
S5 Extras & freeze	14 to 20 Nov	Planned extras (if checkpoint 2 passed), security, bug fixes	**Feature freeze**
S6 Polish	21 to 27 Nov	Bug fixes only, README, API docs, demo data	Ready to submit

Milestones and deliverables

Date	Milestone	Deliverable
16 Oct	Phases 1 to 3 submitted	Planning, literature review and requirements documents
23 Oct	Skeleton live	Deployed website with login
6 Nov	Design submitted, checkpoint 1	Diagrams, database design, wireframes, the end-to-end flow working roughly
13 Nov	Checkpoint 2	All core features working
20 Nov	Feature freeze	No new features after this date
30 Nov	Implementation submitted	Source code, README, deployed link

Date	Milestone	Deliverable
4 Dec	Final presentation	Test reports, user manual, slides, demo video, infographic

Resource allocation

Time (worst case). Every member works about 10 hours a week for 8 weeks, from 9 Oct to 4 Dec.

	Hours
Capacity: 6 members × 10 h × 8 weeks	480
Weekly meetings (1 h each per week)	−48
One member loses a full week	−10
Available	**422**
Core development (estimates × 1.5 safety factor)	167
Planned extras (× 1.5)	59
Code reviews and integration	40
Documentation for all phases (× 1.5)	60
Research, presentation, video, rehearsals (× 1.5)	30
Planned	**356**
Spare	**≈ 66**

The spare hours cover delays and bugs. The extras start only after checkpoint 2, so the core is always finished first.

Role	Development & reviews	Docs & presentation	Total (of 72 h)
R1 Team Leader & DevOps (Ibrahim)	≈ 46 h	≈ 12 h	≈ 58 h
R2 Backend: Matching (Khaled)	≈ 50 h	≈ 8 h	≈ 58 h
R3 Backend: Papers (Mohammed)	≈ 50 h	≈ 8 h	≈ 58 h
R4 Frontend: Donor (Hager)	≈ 48 h	≈ 10 h	≈ 58 h
R5 Frontend: Requester (Zeyad)	≈ 48 h	≈ 10 h	≈ 58 h
R6 UI/UX & QA (Marawan)	≈ 24 h	≈ 42 h	≈ 66 h

Tools and services (cost: $0).

Need	Tool	Plan
Code and tasks	GitHub, GitHub Projects, GitHub Actions	Free
Frontend hosting	Vercel	Free (Hobby)
Backend hosting	Render (Docker)	Free, kept awake with an uptime ping
Database	MongoDB Atlas	Free (M0)
File storage	Cloudinary	Free tier

Need	Tool	Plan
AI paper check	Mock mode for development, free AI tier for the demo with sample papers only	Free
Design	Figma	Free

At launch, the only real cost would be a paid AI model for real papers (roughly $3 to $12 per 1,000 papers).

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3. Task Assignment & Roles

Roles

The team has six roles, and each member takes one. The assignment below is **temporary** and may change once the team confirms it.

#	Role	Member	Main responsibilities	Best fit for someone who…
R1	**Team Leader & DevOps**	Ibrahim Hassan Anwar	Architecture, repo setup, CI, Docker, deployment, authentication and roles, data models, demo data, reviewing pull requests, final say on scope	is the strongest all-round developer and is comfortable with Git and Docker
R2	**Backend: Matching & Alerts**	Khaled Atef Salah	Matching engine and its unit tests, notification waves, push notifications, live updates (Socket.io)	likes logic, algorithms and Node.js
R3	**Backend: Papers & Trust**	Mohammed Eslam Mohammed	AI paper check (`checkPaper()`) and its prompt, trust badges, request states, QR check-in API, admin API (paper review, reports, bans)	likes APIs, databases and working with AI
R4	**Frontend: Donor**	Hager Waleed Shaker	Donor signup and certificate upload, notifications on the phone, accept or decline, QR screen, donation history	likes React and building screens for phones
R5	**Frontend: Requester & Hospital**	Zeyad Mohammed Mohammed	Request form and paper upload, live progress page, partner-hospital portal (confirm papers, scan QR), admin dashboard	likes React, forms and dashboards
R6	**UI/UX & QA**	Marawan Hesham Tolba	Wireframes, UI kit and Arabic RTL guidelines, test plan and bug reports, slides and infographic, lead presenter	has an eye for design, writes clearly and presents well

Document owners

Each phase has one owner who collects the work and submits it. Everyone contributes.

Phase	Owner	Contributors
1. Planning & management	R1	All
2. Literature review (feedback, improvements)	R6	R4
3. Requirements gathering	R6	R4, R5

Phase	Owner	Contributors
4. Analysis & design	R2	R1: architecture, deployment, components R3: ERD, schema, class and state diagrams R4: DFD, activity diagram R5: sequence diagrams R6: wireframes, UI guidelines
5. Implementation (code, README)	R1	All
6. Testing & QA	R6	R2, R3: automated tests R4, R5: UI tests
7. Final presentation & reports	R6	R1: technical docs R4: demo video R5: user manual

How we work

- **Board:** GitHub Projects, with the columns To do, In progress, Review and Done.
- **Meetings:** a 1-hour planning meeting at the start of each sprint, and a short written update every two days.
- **Code:** one branch per feature, a pull request reviewed by one teammate, and CI must pass before merging.
- **Decisions:** after sprint 2, only the team leader can add new features to the plan.

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4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
1	A team member is unavailable for a week or more	Medium	High	Hours planned for the worst case. Two members on each side (backend and frontend) can cover for each other. The API is documented.
2	The work runs out of time	Medium	High	A fixed core scope, checkpoints on 6 and 13 Nov, a feature freeze on 20 Nov and a 3-day buffer.
3	Six people working on the same code cause conflicts or duplicated work	Medium	Medium	Each member owns clear folders and features, pull requests stay small, and there's a short written update every two days.
4	Documentation takes time away from coding	High	Medium	Each phase has one owner, diagrams are written as code in the repo, and documents are written while the feature is being built.
5	The AI marks a fake paper "looks valid"	Medium	Medium	The badge never claims a paper is genuine. Users accept terms saying they are responsible for their papers. Users can report, and abusers are banned. Partner hospitals give the stronger badge.
6	The free AI service is slow or down during the demo	Medium	High	Mock mode (`AI_PROVIDER=mock`), papers checked in advance and a recorded backup video.
7	Push notifications are hard to get working	Medium	Medium	Started early in sprint 3. Live updates on the website are the fallback. Demo on Android Chrome.
8	The free server sleeps and misses notification waves	High	Low	An uptime ping every 5 to 10 minutes.

#	Risk	Likelihood	Impact	Mitigation
9	No real hospital agrees to partner before the demo	High	Low	Demo hospitals have clearly fictional names. We never claim a partnership that doesn't exist.
10	Users' personal data or papers leak	Low	High	Papers are private files, phone numbers are hidden until a donor accepts, no secrets go in the repo, and only sample papers are sent to the free AI. We follow Egypt's Personal Data Protection Law (151/2020).
11	People ask for money or misuse the platform	Medium	High	Clear "no money" rules on every screen, a report button and admin bans.
12	Users see us as responsible for medical outcomes	Low	High	Terms of use say we only connect people. The hospital makes every medical decision.

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5. KPIs (Key Performance Indicators)

These are targets we set for the demo and a small pilot. They are not measured results yet.

Product

KPI	Target	How we measure
Time from request to first donor accepting	Under 30 minutes in the demo	Timestamps stored on each request
Requests that reach at least one accepting donor	≥ 70 % in the pilot	Admin dashboard
Notified donors who accept or decline	≥ 50 %	Notification and response logs
Requests that have a hospital paper	100 %	Enforced by the system
Abuse reports handled by an admin	100 % within 24 hours	Report status and timestamps

Technical

KPI	Target	How we measure
API response time	95 % of requests under 500 ms	Server logs
Uptime during demo week	≥ 99 %	Uptime monitor
Page load on a phone over 4G	Under 3 seconds	Lighthouse
Accessibility score	≥ 90	Lighthouse
Test coverage of core logic (matching, trust badges, request states)	≥ 70 %	Jest coverage report

Adoption (pilot)

KPI	Target	How we measure
Registered donors	50	Database count
Donors with an uploaded certificate	≥ 60 % of registered donors	Database count
Partner hospitals	At least 1 real partner talked to	Outreach log

Project delivery

KPI	Target
Phase documents submitted on time	7 of 7
Core features done by checkpoint 2 (13 Nov)	100 %
Pull requests reviewed before merging	100 %
Critical bugs open at submission	0

Phase 2: Literature Review

Track: Software Development - React Frontend Web Developer\

1. Background

- **Voluntary, unpaid donation is the safest model.** The World Health Organization (WHO) recommends that all blood comes from voluntary, unpaid donors. WHO's 2025 global report says these donors now give more than 85 % of the world's blood, but the share is much lower in low-income countries (63.4 %) [1].
- **Families are often asked to find donors themselves.** A 2017 review of blood transfusion in WHO's Eastern Mediterranean region names Egypt among the countries still "under the weight of family/replacement blood donation" [2]. In practice, the patient's family has to find donors quickly.
- **Social media fills the gap, but badly.** Families post "blood type X needed urgently" on Facebook and WhatsApp. These posts reach people by chance, can't be checked, stay online after the need is over, and publish phone numbers to everyone.
- **Egyptians are willing to donate.** For World Blood Donor Day 2026, a Ministry of Health campaign registered more than 15,000 donors in 12 hours and set a Guinness World Record [3]. The willingness exists. What's missing is a trusted way to connect a specific patient with a nearby donor.

2. Existing solutions

Solution	What it does	Strengths	Gaps
Facebook and WhatsApp posts	Families post urgent requests in groups and on their timelines	Huge reach, free, familiar	No checking, no matching by blood type or distance, phone numbers are public, posts stay up after the need ends
Facebook Blood Donations (in Egypt since June 2020, with the Ministry of Health) [4][5]	Users sign up as donors and get messages from the nearest blood transfusion center	Official partner, large user base	Connects donors to **centers**, not to a specific patient who needs blood now
HOPE / GiveMeHope (Egyptian app by the Hope foundation) [6][7]	Collects blood requests posted on Twitter, lets users post requests, and finds donors with the right blood type within 20 km. Earlier versions also worked by SMS.	Matching by blood type and distance, brings requests together in one place	Built on public posts, and requests aren't checked against hospital papers
Law 3andak Dam (2012) [8]	Matches donors and patients through Twitter and Facebook	One of the first Egyptian attempts	No way to verify donors or requests (noted by the source itself)
University projects (e.g. MSA blood donation app) [9]	Blood bank and donor management apps	Show the need is recognized	Usually built around blood banks, not a community with trust levels

3. The gap BloodLink fills

None of the solutions above combines all of the following. BloodLink does:

Need	How BloodLink answers it
Is the request real?	Every request needs an official hospital paper. Partner hospitals confirm it (**Verified**). Otherwise an AI checks it (**Paper looks valid**).
Is the donor real?	Donors upload an official blood-type certificate and get the same trust badges.
Will the right people see it?	Only compatible, available donors near the hospital are notified, in growing waves.
Is it safe?	Phone numbers stay hidden until a donor accepts, meetings happen only at the hospital, money is never allowed, and there's a report button.
Who is responsible medically?	Always the hospital. BloodLink only connects people.

4. What we learned from them

Solution	Lesson	How BloodLink uses it
Facebook and WhatsApp posts	People already share requests on social media, and that reach is valuable	A WhatsApp share card that links back to the checked request
Facebook Blood Donations	Working with official bodies builds trust	Partner hospitals confirm papers and give the **Verified** badge
HOPE / GiveMeHope	Matching by blood type and distance works	Matching by compatibility, distance waves (5 km, then 15 km) and trust level
HOPE / GiveMeHope	SMS reaches people without smartphones	SMS alerts are planned as a future improvement
Law 3andak Dam	Its main weakness was that nothing could be verified	Every request needs an official hospital paper
University projects	They focus on blood banks	BloodLink connects a specific patient with nearby donors

5. Suggested Improvements

Improvements we have already planned, and ideas for after the course.

#	Improvement	Why	When
1	English version of the interface	Reach foreigners living in Egypt and the evaluators	Sprint 5 (planned extra)
2	Map view of nearby requests	Easier to judge distance than a list	Sprint 5 (planned extra)
3	WhatsApp share card for a request	People already use WhatsApp; the card links back to the checked request	Sprint 5 (planned extra)

#	Improvement	Why	When
4	Donation history and simple charts	Motivates donors, shows impact	Sprint 5 (planned extra)
5	Signing real partner hospitals	More requests and donors get the stronger "Verified" badge	After the course
6	SMS alerts for donors without smartphones	Reach more donors	After the course (paid service)
7	Blood stock levels at partner hospitals	Donors can help before an emergency	After the course
8	Mobile app (Android and iPhone), built on the same backend	More reliable notifications, especially on iPhone	Future, not part of this project

6. References

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Phase 3: Requirements Gathering

Track: Software Development - React Frontend Web Developer\

1. Stakeholder Analysis

Stakeholder	Who they are	What they need	Influence	How we involve them
Requester	A patient's family member or friend	Find compatible donors fast, without posting everywhere	High	Interviews, usability test of the request flow
Donor	A volunteer who wants to give blood	Only relevant, real requests close to home; privacy	High	Short survey, usability test of the donor flow
Partner hospital	Hospital staff using our portal	Confirm papers quickly, record donations, reach donors for their patients	High	Talk to 1 or 2 hospitals; demo uses fictional ones
Non-partner hospital	Any other official hospital	Nothing from us; it still examines the donor and decides	Low	None. Its papers are checked by AI
Admin	Our moderation team	Review unclear papers, handle reports, ban abusers	Medium	Built by the team
AI service provider	The vision AI API	Clear prompts, sample papers only on the free tier	Low	Through one `checkPaper()` function
DEPI lecturer and evaluators	Grade the project	Clear docs, working demo, good code	High	Phase reviews and the final presentation
Project team	The 6 members	A realistic plan and clear roles	High	Weekly meetings, GitHub board

2. User Stories

Format: *As a [role], I want [goal], so that [benefit].*

Donor

ID	Story	Acceptance criteria	Priority
US-01	As a donor, I want to sign up with my blood type and area, so that I only get requests that fit me	Signup asks for name, phone, blood type, governorate, area and location. The user must accept the terms of use.	Must
US-02	As a donor, I want to upload my hospital blood-type certificate, so that people can trust me	Accepts JPG, PNG or PDF up to 5 MB. Shows the resulting badge: **Verified**, **Paper looks valid** or **Needs review**.	Must
US-03	As a donor, I want to be notified about requests near me, so that I can help in time	A push notification arrives only when the blood type is compatible, I'm available, and the hospital is within the wave radius.	Must
US-04	As a donor, I want to accept or decline a request, so that the requester knows quickly	One tap to accept or decline. On accept, I see the hospital address and the requester's phone number.	Must

ID	Story	Acceptance criteria	Priority
US-05	As a donor, I want a QR code when the hospital is a partner, so that the staff can record my donation	QR shows only for partner hospitals. At other hospitals, I see "No QR here, confirm after donating."	Must
US-06	As a donor, I want to turn my availability off, so that I'm not disturbed	A visible on/off switch. No notifications while it's off.	Must
US-07	As a donor, I want to see my donation history, so that I feel the impact	List of past donations with date, hospital and how each was confirmed	Should

Requester

ID	Story	Acceptance criteria	Priority
US-08	As a requester, I want to create a request with the hospital paper, so that donors trust it	The request can't be sent without a paper. Asks for blood type, units, hospital, urgency and deadline.	Must
US-09	As a requester, I want to see the badge my request got, so that I know how it looks to donors	Shows **Verified** (partner confirmed), **Paper looks valid** (AI) or **Needs review**	Must
US-10	As a requester, I want to follow the progress live, so that I don't keep calling people	"2 of 3 units pledged" updates without refreshing the page	Must
US-11	As a requester, I want to see a donor's phone number only after they accept, so that contact is safe for both sides	No phone numbers appear before acceptance	Must
US-12	As a requester at a non-partner hospital, I want to confirm the donation myself, so that the request can be closed	Both requester and donor confirm. The donor's badge does **not** change.	Must
US-13	As a requester, I want to report a suspicious donor or message, so that abusers are removed	Report button on every profile and request, with a reason	Must
US-14	As a requester, I want to share my request on WhatsApp, so that more people see it	Share card with a link back to the checked request	Should

Partner hospital

ID	Story	Acceptance criteria	Priority
US-15	As partner-hospital staff, I want to confirm papers issued by my hospital, so that requests and donors get the **Verified** badge	List of pending papers with Confirm and Reject buttons	Must
US-16	As partner-hospital staff, I want to scan a donor's QR code, so that I can record what happened	Outcome: **Donated**, **Not able to donate today (medical decision)** or **Didn't show up**	Must

Admin

ID	Story	Acceptance criteria	Priority
US-17	As an admin, I want to review papers the AI marked as unclear, so that no user is stuck	Queue of unclear papers with Approve, Ask for a clearer photo and Reject	Must

ID	Story	Acceptance criteria	Priority
US-18	As an admin, I want to handle reports and ban accounts, so that the community stays safe	Reports list with Dismiss, Warn and Ban. Banned users can't log in.	Must
US-19	As an admin, I want to approve partner hospitals, so that only real hospitals can confirm papers	Partner accounts are inactive until an admin approves them	Must
US-20	As an admin, I want to see key numbers, so that I know if the platform works	Cards: active requests, donors, average time to first acceptance, open reports	Must

3. Use Cases

UC-01: Create a blood request

Item	Details
Actor	Requester
Precondition	Logged in, terms accepted
Main flow	1. Fill in blood type, units, hospital, urgency and deadline. 2. Upload the hospital paper. 3. Review the "check your answers" page. 4. Send. 5. The system checks the paper: partner hospital, so it goes to the hospital to confirm; non-partner, so the AI checks it. 6. The request becomes active with a badge and matching starts.
Alternative flows	5a. The AI finds the paper unclear: the request waits in the admin review queue. 5b. The partner hospital rejects it: the request is rejected and the requester is told why.
Postcondition	An active request with a trust badge, and the first wave of notifications sent

UC-02: Notify donors and accept

Item	Details
Actor	System, Donor
Precondition	An active request
Main flow	1. The system finds compatible, available donors within 5 km and ranks them (distance, same type, trust, reliability). 2. It sends push notifications. 3. A donor opens the request and taps **I can donate**. 4. The pledge count goes up live for the requester. 5. Phone numbers are shown to both.
Alternative flows	1a. Not enough donors accept: the next wave widens to 15 km. 3a. The donor declines: the system notifies the next donor. 3b. Nobody accepts before the deadline: the request expires.
Postcondition	A pledge linked to the request

UC-03: Record a donation at a partner hospital

Item	Details
Actor	Donor, Partner-hospital staff
Precondition	The donor accepted a request at a partner hospital
Main flow	1. The donor shows the QR code at the hospital. 2. Staff scan it in the portal. 3. The hospital examines the donor and decides. 4. Staff record **Donated**. 5. The donor becomes **Verified** and is paused from new requests for a while. 6. The request count updates.

Item	Details
Alternative flows	4a. **Not able to donate today:** the pledge is released and the next wave may start. 4b. **Didn't show up:** the donor's reliability score goes down.
Postcondition	A recorded donation, confirmed by the hospital

UC-04: Confirm a donation at a non-partner hospital

Item	Details
Actor	Donor, Requester
Precondition	The donor accepted a request at a non-partner hospital
Main flow	1. They meet at the hospital, which decides medically. 2. The donor taps **I donated**. 3. The requester taps **Confirm**. 4. The donation is recorded as self-confirmed.
Alternative flows	3a. The requester says it didn't happen: the case goes to an admin.
Postcondition	A self-confirmed donation. The donor's badge stays the same.

UC-05: Handle a report

Item	Details
Actor	Any user, Admin
Main flow	1. A user reports a profile or request with a reason, for example asking for money. 2. The admin reviews it. 3. The admin dismisses it, warns the user or bans the account. 4. The reporter is told the result.
Postcondition	The report is closed and the action is logged

4. Functional Requirements

ID	Requirement	Priority
Accounts		
FR-01	Users can register and log in with email or phone and a password	Must
FR-02	The system supports four roles: donor, requester, partner hospital, admin. A user can be both donor and requester.	Must
FR-03	Users must accept the terms of use (the site only connects people) at signup	Must
FR-04	Partner-hospital accounts stay inactive until an admin approves them	Must
Papers and trust		
FR-05	Donors upload an official hospital blood-type certificate	Must
FR-06	Requesters must upload an official hospital paper before a request is published	Must
FR-07	Papers from partner hospitals are sent to that hospital to confirm, and give the **Verified** badge	Must
FR-08	Papers from other hospitals are checked by AI, and give the **Paper looks valid** badge	Must
FR-09	The AI check confirms the paper looks official, the name and blood type match, and the date is recent. Unclear results go to the admin queue.	Must

ID	Requirement	Priority
FR-10	The AI never labels a paper as genuine. The badge text is fixed.	Must
FR-11	Uploaded papers are private and never shown to other users	Must
Matching and notifications		
FR-12	Matching uses blood compatibility, availability, pause period, distance and trust level	Must
FR-13	Notifications go out in waves: 5 km first, then 15 km	Must
FR-14	Donors get browser push notifications and a list of requests on the website	Must
FR-15	Requests expire automatically after their deadline	Must
Response and meeting		
FR-16	Donors can accept or decline a request	Must
FR-17	Phone numbers are shown only after the donor accepts	Must
FR-18	Requesters see the pledge count live	Must
FR-19	At partner hospitals, the donor gets a QR code, and staff scan it to record **Donated**, **Not able to donate today** or **Didn't show up**	Must
FR-20	At non-partner hospitals, there is no QR code. Donor and requester confirm the donation themselves. This never upgrades the donor's badge.	Must
FR-21	After a donation, the donor is paused from new requests for a period (general guidance; the hospital decides)	Must
Safety and admin		
FR-22	Every profile and request has a report button	Must
FR-23	Admins can review unclear papers, handle reports and ban accounts	Must
FR-24	The admin dashboard shows key numbers: active requests, donors, time to first acceptance, open reports	Must
Extras		
FR-25	English version of the interface	Should
FR-26	Map view of requests	Should
FR-27	WhatsApp share card	Should
FR-28	Donation history and charts	Should
FR-29	SMS alerts, blood stock at partner hospitals	Could

5. Non-functional Requirements

ID	Category	Requirement	How we check it
NFR-01	Performance	95 % of API requests answer in under 500 ms	Server logs
NFR-02	Performance	Pages load in under 3 seconds on a phone over 4G	Lighthouse
NFR-03	Performance	Matching for one request finishes in under 2 seconds	Unit test with demo data

ID	Category	Requirement	How we check it
NFR-04	Security	Passwords are hashed with bcrypt. Login uses a JWT in an httpOnly cookie.	Code review
NFR-05	Security	Every input is validated on the server (zod). Login and upload are rate-limited.	Code review, API tests
NFR-06	Security	No secrets in the repo; all keys live in `.env`	CI check, `.env.example`
NFR-07	Privacy	Phone numbers and papers are shared only as described in FR-11 and FR-17. Users consent before the AI check. We follow Egypt's Personal Data Protection Law (151/2020).	Code review, test cases
NFR-08	Privacy	Only sample papers are sent to the free AI tier	Team rule, checklist
NFR-09	Usability	Arabic first, right-to-left layout, works on phones from 360 px wide	Manual test on phones
NFR-10	Usability	A first-time requester can send a request in under 3 minutes	Usability test
NFR-11	Accessibility	Lighthouse accessibility score ≥ 90, visible keyboard focus, readable contrast	Lighthouse
NFR-12	Reliability	Uptime ≥ 99 % during demo week	Uptime monitor
NFR-13	Reliability	If the AI service fails, the website switches to mock mode or the review queue, never to an error page	Test with the AI turned off
NFR-14	Maintainability	TypeScript on both sides, ESLint and Prettier, small reviewed pull requests	CI
NFR-15	Testability	≥ 70 % test coverage of core logic (matching, trust badges, request states)	Jest coverage report
NFR-16	Portability	The whole website runs locally with `docker compose up`	Fresh-laptop test
NFR-17	Cost	The demo runs at $0 on free tiers	Monthly check of all accounts